Andrei-Cătălin **Popa**

PhD Student in Astrophysics

Lennard-Jones School of Chemical and Physical Sciences, Keele University, Staffordshire, ST5 5BG, UK

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Education

Keele University *Keele, UK*

PhD in Astrophysics *Oct. 2026 – present*

Supervisor: Dr. Pierre Maxted.

Thesis: *Calibrating Stellar Models with Chemical Tracers of Internal Mixing Processes.*

PLATO (launch 2026) aims to discover and characterise rocky planets orbiting Sun-like stars, with stellar ages measured to 10% accuracy. Developing a new method to calibrate next-generation stellar models using chemical tracers in binary star spectra to probe the efficiency of internal mixing processes (convective overshooting, atomic diffusion, and rotation) in Sun-like stars of known mass.

Université Côte d'Azur *Nice, France*

MSc in Astrophysics and Space Science (Erasmus Mundus Joint Master's Programme) *Mar. 2025 – Sept. 2026*

Thesis: *Characterising M-dwarf Stars as Exoplanet Hosts in Preparation for the PLATO Mission.*

Advisors: Dr. Orlagh Creevey (OCA), Dr. Valentina D'Orazi (Tor Vergata), Dr. Ulrike Heiter (Uppsala).

Research Internship: *Fundamental Parameters of FGK Stars.*

Advisors: Dr. Orlagh Creevey (OCA) and Dr. Caroline Soubiran (Bordeaux).

Università degli Studi di Roma "Tor Vergata" *Rome, Italy*

MSc in Astrophysics and Space Science (Erasmus Mundus Joint Master's Programme) — First Semester *Sept. 2024 – Mar. 2025*

Consortium of four universities: Università degli Studi di Roma "Tor Vergata", Universität Bremen, Université Côte d'Azur, and University of Belgrade.

University College London *London, United Kingdom*

MSci in Physics (First Class Honours) *Sept. 2018 – Jun. 2022*

Supervisor: Prof. Daisuke Kawata (Mullard Space Science Laboratory).

Thesis: *Hunting for Intermediate-Mass Black Hole Binaries.*

Research Experience

Astrophysics Research Centre *Keele, United Kingdom*

PhD Research: Calibrating Stellar Models with Chemical Tracers of Internal Mixing Processes *Oct. 2026 – present*

- Supervised by Dr. Pierre Maxted. Research within the ESA *PLATO* mission consortium, focused on using chemical abundance tracers in binary stars to calibrate stellar structure and evolution models.
- Investigating internal mixing processes (convective overshooting, atomic diffusion, rotation) through precision constraints provided by binary star systems and *PLATO* lightcurves.

Observatoire de la Côte d'Azur *Nice, France*

Master's Thesis: Characterising M-dwarf Stars as Exoplanet Hosts – Preparation for the PLATO Mission *Jan. 2026 – Sept. 2026*

- Supervised by Dr. Orlagh Creevey (OCA), Dr. Ulrike Heiter (Uppsala University) and Dr. Valentina D'Orazi (Tor Vergata), within the Erasmus Mundus Joint Master's in Astrophysics and Space Science (MASS).
- Deriving luminosities and radii of benchmark M-dwarf stars via spectral energy distribution (SED) fitting and bolometric flux calculations, using *Gaia* XP spectra and multi-band photometry.
- Building and testing Python pipelines for photometric and spectroscopic data comparison, integrating literature stellar parameters for cross-validation.
- Contributing to *PLATO* host-star modelling by refining fundamental parameters used in the mission's stellar characterisation framework.

Astrophysics Research Institute *Liverpool, United Kingdom*

Summer Research Project: White Dwarf Cooling Sequences and Infrared Colour Excess *Jul. 2025 – Sept. 2025*

- Supervised by Prof. Maurizio Salaris. Investigated whether the apparent broadening of the white dwarf cooling sequence in NGC 6397 reflects an intrinsic IR-excess population or statistical scatter.
- Analysed JWST and HST photometric data; constructed synthetic white dwarf populations from theoretical isochrones and a Salpeter IMF, with Monte Carlo uncertainty propagation.
- Developed Python routines to compare observed and simulated colour–magnitude diagrams, applying Gaussian Mixture Models (GMM) and Bayesian Information Criterion (BIC) analysis.

Scholarships & Honours

2026–2030	**STFC PhD Studentship**, Science and Technology Facilities Council / UKRI	*£21,805 p.a.*
2024–2026	**Erasmus Mundus (EMJM) Scholarship**, European Education and Culture Executive Agency	*€33,600*
2022	**Summer Research Internship Scholarship**, Dept. of Physics & Astronomy, UCL	*£2,000*
2022	**"Teach Physics" Internship Award**, Ogden Trust — Eastbury Community School, London	*£1,400*

Teaching

Eastbury Community School *Barking, United Kingdom*
SUMMER PHYSICS TEACHING INTERN — OGDEN TRUST "TEACH PHYSICS" PROGRAMME *Jun. – Jul. 2022*
Taught A-level and GCSE physics through interactive sessions and lab exercises; created classroom materials aligned with UK physics curricula; supported Science Club activities.

Outreach

UCL Observatory *Mill Hill, United Kingdom*
DEMONSTRATOR *Oct. 2019 – Sept. 2022*
Trained to operate robotic and manual telescopes (14-inch Schmidt–Cassegrain; twin 24/18-inch refractors). Led public observing sessions for school groups and members of the public; developed science communication skills across diverse audiences.

Department of Physics & Astronomy, University College London *London, United Kingdom*
OUTREACH INTERN *Jun. – Sept. 2022*
Organised outreach activities in local schools and community groups; planned and delivered science communication events including UCL's public festival *Your Universe*; produced materials translating astrophysical research for non-specialist audiences.

Service & Committees

2024–2025	**Student Representative**, Quality Evaluation Committee, Erasmus Mundus MASS Programme	*Rome/Nice*

Skills & Professional Memberships

Programming	Python, MATLAB, IDL
Astronomical Software	TOPCAT, SAO DS9, AstroImageJ, VOSA, iSpec
Tools & Environments	Git, LaTeX, UNIX/Linux
Professional Memberships	**Institute of Physics (IOP)**; **Mensa Romania**; **England Handball Referee Association**
Languages	**English:** Fluent | **Romanian:** Native | **French:** Intermediate

References

Dr Pierre Maxted *Keele University*
READER IN ASTROPHYSICS
contact: p.maxted@keele.ac.uk

Dr Orlagh Creevey *Observatoire de la Côte d'Azur*
RESEARCH ASTRONOMER
contact: orlagh.creevey@oca.eu

Dr Valentina D'Orazi *Università di Roma "Tor Vergata"*
TENURE-TRACK RESEARCHER IN ASTROPHYSICS
contact: vdorazi@roma2.infn.it